

HYDROGENATED NBR: LOW-MOLECULAR-WEIGHT AND FUNCTIONALIZED HNBR PREPARATION PROCESSES

1. ABSTRACT

This report synthesizes three primary patents addressing hydrogenated nitrile rubber (HNBR) preparation, with emphasis on lowering molecular weight, controlling residual unsaturation, and improving processability while preserving the characteristic heat, ozone, and oil resistance of HNBR. The target material is hydrogenated or partially hydrogenated nitrile-butadiene rubber derived from NBR precursors, including conventional NBR and PEG-acrylate-functionalized NBR. Across the selected patents, two major process families emerge: (1) simultaneous metathesis/hydrogenation of nitrile rubber in the presence of hydrogen and ruthenium- or osmium-based catalysts, and (2) latex hydrogenation of functionalized NBR followed by coagulation to obtain processable HNBR copolymers. The first approach is represented by US7951875B2 and EP2473281B1, both focused on reducing molecular weight and narrowing molecular weight distribution during hydrogenation, with EP2473281B1 explicitly claiming simultaneous metathesis and hydrogenation in hydrogen atmosphere. The second approach, US20220073666A1, targets PEG acrylate-HNBR copolymers and emphasizes latex hydrogenation of PEG acrylate-NBR latex, followed by coagulation, to retain functionalization while improving processing behavior. The patents collectively show that HNBR synthesis is not limited to simple post-hydrogenation of NBR; instead, process design can integrate chain scission via metathesis, catalyst selection to avoid nitrile reduction, and latex-state hydrogenation to manage viscosity and downstream handling. Key formulation/process trends include use of co-olefins such as ethylene or 1-hexene in metathesis-assisted molecular-weight reduction, ruthenium catalysts with phosphine or N-heterocyclic carbene ligation, hydrogen pressures in the multi-megapascal range for latex hydrogenation, and monomer windows spanning acrylonitrile, conjugated dienes, and PEG acrylate comonomers. Overall, the evidence supports HNBR synthesis strategies that balance hydrogenation degree, residual double-bond content, and molecular weight control to achieve improved processability without sacrificing the performance profile that makes HNBR valuable in seals, hoses, and oilfield elastomers.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight hydrogenated nitrile rubber prepared by a metathesis-based molecular-weight reduction approach applied to nitrile rubber feedstock, followed by hydrogenation-related processing described in the context of HNBR. The examples show use of a cement solution, degassing with ethylene or nitrogen, and reaction at elevated temperature with catalyst loading to reduce molecular weight and Mooney viscosity. - Rubber charge: 200 g — Example 1 - MCB solvent: 1133 g — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber charge: 200 g — Example 2 - MCB solvent: 1133 g — Example 2 - Solid content: 15 wt.-% — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Rubber charge: 75 g — Example 3 - MCB solvent: 1175 g — Example 3 - Solid content: 6 wt.-% — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - Rubber charge: 75 g — Example 4 - MCB solvent: 1175 g — Example 4 - Solid content: 6 wt.-% — Example 4 - 1-hexene amount: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Co-olefin pressure: 500 psi — Table 1 - Co-olefin pressure: 500 psi — Table 1 - Co-olefin pressure: 500 psi — Table 1 - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - MWD: 2.0 — typical product - Mn: 85 kg/mol — starting polymer - Mw: 296 kg/mol — starting polymer - MWD: 3.4 — starting substrate feedstock - Mooney viscosity: 73 — Therban® A3407 (comp.) - Mn: 73,000 — Experiment 1 - Mw: 189,000 — Experiment 1 - MZ: 441,000 — Experiment 1 - PDI: 2.59 — Experiment 1 - Mooney viscosity: 43 — Experiment 1 - Mn: 60,000 — Experiment 2 - Mw: 136,000 — Experiment 2 - MZ: 277,000 — Experiment 2 - PDI: 2.27 — Experiment 2 - Mooney viscosity: 28 — Experiment 2 - Mn: 31,000 — Experiment 3 - Mw: 59,000 — Experiment 3 - MZ: 98,000 — Experiment 3 - PDI: 1.90 — Experiment 3 - Mooney viscosity: 3 — Experiment 3 - Mn: 55,000 — Experiment 4 - Mw: 111,000 — Experiment 4 - MZ: 1,197,000 — Experiment 4 - PDI: 2.02 — Experiment 4 - Mooney viscosity: 31 — Experiment 4

Relevant Experimental Evidence - Example 1: Rubber cement was degassed three times with ethylene at 6.9×10^5 Pa, establishing the metathesis/hydrogenation reactor setup. - Example 2: Repeated ethylene degassing under the same pressure confirmed reproducibility of the process conditions. - Example 3: Lower-solids cement (6 wt.-%) was processed under ethylene, showing that feed concentration affects product molecular weight and viscosity. - Example 4: Use of

1-hexene co-olefin at 80° C. with catalyst loading demonstrated strong molecular-weight reduction and narrowing of MWD, with Mooney viscosity reduced to 31 in the table summary.

Technical Relevance This patent is directly relevant because it targets the core HNBR problem of excessive molecular weight and viscosity after hydrogenation. It shows that metathesis-assisted chain scission can be used to generate lower-MW HNBR-like materials with improved processability.

Patent 2

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low-molecular-weight hydrogenated nitrile rubber by simultaneous metathesis and hydrogenation of nitrile rubber in the presence of hydrogen. The process uses ruthenium- or osmium-based metal compounds and optionally a co-olefin, aiming to reduce molecular weight and residual double-bond content in a single step. - Catalyst amount: 0.041 phr — Example 1 (HNBR 1) - Catalyst amount: 0.045 phr — Example 1 (HNBR 2) - Catalyst amount: 0.060 phr — Example 1 (HNBR 3) - Metal content: 65.6 ppm — Example 1 - Co-catalyst: NIL — Example 1 (HNBR 1) - Co-catalyst: NIL — Example 1 (HNBR 2) - Co-catalyst: Triphenylphosphine — Example 1 (HNBR 3) - Co-catalyst amount: 1.0 phr — Example 1 (HNBR 3) - Reaction time: 0 min — Example 1 - RDB: 100% — Example 1 (all HNBR samples at 0 min) - Reaction time: 60 min — Example 1 - RDB: 2.5% — Example 1 (HNBR 1 at 60 min) - RDB: 7.0% — Example 1 (HNBR 2 at 60 min) - RDB: 15.4% — Example 1 (HNBR 3 at 60 min) - Reaction time: 120 min — Example 1 - RDB: 0.6% — Example 1 (HNBR 1 at 120 min) - RDB: 4.5% — Example 1 (HNBR 2 at 120 min) - RDB: 3.1% — Example 1 (HNBR 3 at 120 min) - Reaction time: 180 min — Example 1 - RDB: 0.5% — Example 1 (HNBR 1 at 180 min) - RDB: 3.9% — Example 1 (HNBR 2 at 180 min) - RDB: 1.3% — Example 1 (HNBR 3 at 180 min) - Reaction time: 240 min — Example 1 - RDB: 0.4% — Example 1 (HNBR 1 at 240 min) - RDB: 3.5% — Example 1 (HNBR 2 at 240 min) - RDB: 0.7% — Example 1 (HNBR 3 at 240 min) - Mw: 113371 g/mol — Example 1 (HNBR 1) - Mn: 48352 g/mol — Example 1 (HNBR 1) - PDI: 2.34 — Example 1 (HNBR 1) - Mooney viscosity: 33.3 — Example 1 (HNBR 1) - Mw: 133758 g/mol — Example 1 (HNBR 2) - Mn: 54595 g/mol — Example 1 (HNBR 2) - PDI: 2.45 — Example 1 (HNBR 2) - Mooney viscosity: 41.3 — Example 1 (HNBR 2) - Mw: 195304 g/mol — Example 1 (HNBR 3) - Mn: 64130 g/mol — Example 1 (HNBR 3) - PDI: 3.05 — Example 1 (HNBR 3) - Mooney viscosity: 69.3 — Example 1 (HNBR 3)

Relevant Experimental Evidence - Example 1: Compared three catalyst systems and showed that the inventive catalyst (HNBR 1) achieved the fastest and deepest hydrogenation, with RDB reduced to 0.4% at 240 min. - Example 1: The inventive catalyst also produced the lowest molecular weight and lowest Mooney viscosity among the compared samples. - Example 1: The industrial-norm catalyst (HNBR 3) gave higher residual double-bond content and much higher Mooney viscosity, demonstrating the benefit of the disclosed catalyst system.

Technical Relevance This patent is highly relevant because it explicitly teaches simultaneous metathesis and hydrogenation of nitrile rubber in hydrogen, which is a direct route to low-residual-unsaturation HNBR. It provides concrete catalyst-loading and performance data linking catalyst choice to molecular-weight reduction and hydrogenation efficiency.

Patent 3

- Patent Number: US20220073666A1
- Patent Title: Process for producing peg acrylate-hnbr copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses production of PEG acrylate-HNBR copolymer by hydrogenating a PEG acrylate-NBR latex and then coagulating the resulting latex. The process is positioned as a latex-hydrogenation route for functionalized NBR, preserving PEG acrylate functionality while converting the diene double bonds to a hydrogenated HNBR structure. - PEG acrylate-NBR latex nitrile unit content: 10% to 60% by weight — Claim 1 - PEG acrylate-NBR latex diene unit content: 20% to 80% by weight — Claim 1 - PEG acrylate unit content: 10% to 60% by weight — Claim 1 - R: hydrogen or branched or unbranched C1-C20-alkyl — Claim 1 - n: 1 to 8 — Claim 1 - R1: hydrogen or methyl — Claim 1 - Nitrile unit content: 15% to 40% by weight — Claim 2 - Diene unit content: 30% to 65% by weight — Claim 3 - PEG acrylate unit content: 10% to 50% by weight — Claim 4 - Nitrile unit content: 15% to 35% by weight — Claim 5 - Diene unit content: 35% to 60% by weight — Claim 5 - PEG acrylate unit content: 20% to 50% by weight — Claim 5 - Reaction temperature: 80° C. to 160° C. — prior-art latex hydrogenation conditions - Hydrogen pressure: 3 000 000 to 8 000 000 pascals — prior-art latex hydrogenation conditions

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it extends HNBR synthesis to functionalized PEG acrylate-containing copolymers and uses latex hydrogenation, a process route important for processability. It also defines monomer windows that can be used to tune the balance between hydrogenation, functionality, and rubber performance.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7951875B2 and EP2473281B1 both focus on lowering molecular weight and Mooney viscosity during HNBR-related processing, but US7951875B2 emphasizes metathesis-assisted molecular-weight reduction with co-olefins such as ethylene and 1-hexene, whereas EP2473281B1 emphasizes simultaneous metathesis and hydrogenation in hydrogen atmosphere.
- EP2473281B1 provides the clearest hydrogenation-performance data, with residual double bond content reduced to 0.4% at 240 min for the inventive catalyst, while US7951875B2 reports strong molecular-weight reduction and PDI narrowing but does not present the same explicit RDB time profile.
- US20220073666A1 differs from the other two patents by focusing on PEG acrylate-functionalized NBR latex and a hydrogenation-then-coagulation route rather than metathesis-based chain scission.
- Catalyst systems differ substantially: EP2473281B1 uses ruthenium/osmium compounds in hydrogen, US7951875B2 relies on metathesis catalysts with co-olefins, and US20220073666A1 cites group 8/9/10 transition-metal catalysts for latex hydrogenation in prior-art context.
- Process state also differs: US20220073666A1 is latex-based, while US7951875B2 and EP2473281B1 operate on dissolved/cement or solution-phase nitrile rubber.
- Temperature and pressure disclosure is strongest in US20220073666A1 for latex hydrogenation context (80° C. to 160° C., 3 000 000 to 8 000 000 Pa), while US7951875B2 explicitly uses 80° C. and 500 psi co-olefin pressure in the metathesis examples.

4. CONCLUSION

The evidence supports three complementary HNBR synthesis strategies: simultaneous metathesis/hydrogenation for direct molecular-weight reduction, metathesis-assisted chain scission with co-olefins, and latex hydrogenation of functionalized NBR followed by coagulation. For low-MW HNBR, the strongest evidence favors ruthenium-based simultaneous metathesis/hydrogenation under hydrogen with catalyst loadings in the 0.041-0.060 phr range and reaction times up to 240 min, which can drive RDB below 1% and reduce Mooney viscosity substantially. For processability-focused HNBR, latex hydrogenation at 80-160° C. and 3-8 MPa hydrogen pressure is a practical route, especially when preserving PEG acrylate functionality. Overall, the recommended synthesis direction from the evidence is to combine controlled hydrogenation with either metathesis-based molecular-weight reduction or latex-state processing, while targeting low residual unsaturation, moderate catalyst loading, and monomer compositions that maintain nitrile content and diene-derived performance.

5. REFERENCES

- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US20220073666A1 | Process for producing peg acrylate-hnbr copolymer | Not disclosed | US | 2022 | <https://patents.google.com/patent/US20220073666A1/en>